

ReconAgent Intern Assignment

Build a Specialist Security Agent

Program: Entersoft ReconAgent | **Issued:** 30 June 2026

Dear Intern,

Welcome to the ReconAgent intern project. You and your teammate will build **one specialist security agent** — a small service that accepts a **natural-language prompt**, runs reconnaissance tools, and returns a **structured JSON response**. You are not rebuilding the full orchestrator; you are making your agent an expert at one phase of web application recon.

Your team assignment, tools, API port, and deadlines are listed in this document. Read it fully before you start. Contact your instructor if your team assignment looks incorrect.

1. Your Team Assignments

Team	Agent	Students	Port	Agent ID
Team 1	Scout	ANSH VARMA AVIRAL YADAV	8001	agent-scout
Team 2	Mapper	ISHAAN AGARWAL NITYA SRI RAVURI	8002	agent-mapper
Team 3	Analyst	PRANEET RAJ LINGAMALLU RITHIKA PERI	8003	agent-analyst
Team 4	Prober	RONIL BORAH SAI KEERTHI CHITLA	8004	agent-prober
Team 5	Striker	SAI SIDDHARTH DAVULURI SAMANTULA DEVENDRA SUPREETH	8005	agent-striker

Shared tool for all teams: **httpx** (live target verification). Each agent also accepts optional **context** JSON from upstream agents.

2. What You Must Build

- A FastAPI service running on your assigned port.
- Endpoint **POST /agents/{agent_id}/tasks** — send a prompt, receive a response.
- Endpoint **GET /health** — returns agent status and your tool allowlist.
- Tool allowlist enforced — your agent may only call its assigned tools (+ httpx).
- One POST = one prompt = one response. Do not expose internal steps in the API.

3. API Contract

Request (POST body)

```
{
  "prompt": "Your natural-language task here.",
  "target": "https://target.example.com",
  "context": {}
}
```

prompt (required) — what the agent must do. **target** (required) — URL or hostname in scope. **context** (optional) — JSON from upstream agents in the pipeline.

Response (200 OK)

```
{
  "agent_id": "agent-scout",
  "status": "completed",
  "response": { "summary": "...", "findings": [] }
}
```

On failure return status **failed** with an **error** message. Timeout per request: **5 minutes**.

Example curl (replace port and agent_id with yours)

```
curl -X POST "http://localhost:8001/agents/agent-scout/tasks" ^
-H "Content-Type: application/json" ^
-d "{\"prompt\": \"Is the target live and what tech does it
use?\", \"target\": \"https://example.com\", \"context\": {}}"
```

4. Team Briefs — Tools & Sample Prompts

Team 1: Scout (F1 Fingerprinting)

Field	Value
Students	ANSH VARMA, AVIRAL YADAV
Agent ID	agent-scout
Port	8001
Goal	Find, confirm, and profile in-scope web assets.
Tools	httpx, wafw00f, whatweb, shcheck, testssl, nmap, wpscan (+ httpx)
Sample prompt	Fingerprint https://target.example.com — is it live, what tech stack, WAF, and TLS posture?

Team 2: Mapper (F2 Surface Enumeration)

Field	Value
Students	ISHAAN AGARWAL, NITYA SRI RAVURI
Agent ID	agent-mapper
Port	8002
Goal	Map attack surface — endpoints, JS, ports, hidden paths.
Tools	naabu, katana, gau, waybackurls, dirsearch, jsluice, nirjas, source_maps_downloader, nmap_service (+ httpx)
Sample prompt	Map all routes, JS files, API paths, and open ports for https://target.example.com. Use Scout context if provided.

Team 3: Analyst (F3 Passive Scan)

Field	Value
Students	PRANEET RAJ LINGAMALLU, RITHIKA PERI
Agent ID	agent-analyst
Port	8003
Goal	Passive analysis — secrets and misconfigs, no intrusive payloads.
Tools	trufflehog, secretfinder, linkfinder, gitleaks, git_secrets, mapextractor, jshole, nuclei_passive, httpx_enrichment
Sample prompt	Analyze the endpoints and JS from Mapper context. Find exposed secrets and passive vulnerability indicators.

Team 4: Prober (F4 Active Scan (safe))

Field	Value
Students	RONIL BORAH, SAI KEERTHI CHITLA

Field	Value
Agent ID	agent-prober
Port	8004
Goal	Active scanning without mandatory HITL tools.
Tools	nuclei_active, testssl_deep, ffuf, wfuzz, kxss, corsy, nikto, wpscan_full, droopescan (+ httpx)
Sample prompt	Run active scans on https://target.example.com using upstream context. Report confirmed issues and what S

Team 5: Striker (F4 Active Scan (exploit))

Field	Value
Students	SAI SIDDHARTH DAVULURI, SAMANTULA DEVENDRA SUPREETH
Agent ID	agent-striker
Port	8005
Goal	Validate high-impact vulns with targeted payloads (mock HITL required).
Tools	sqlmap, xssstrike, dalfox, smuggler, ssrfmap, tplmap, crlfuzzer (+ httpx)
Sample prompt	Validate SQLi/XSS/SSRF on the injection points from Prober context. Require mock HITL approval before ea

5. Pipeline (Week 3 — Optional)

Scout → Mapper → Analyst → Prober → Striker. Pass each agent's **response** JSON as **context** to the next agent.

6. Implementation

Reuse existing tool wrappers from the ReconAgent codebase:

```
from tools import TOOL_REGISTRY
result = TOOL_REGISTRY['httpx']().run(target)
```

Repository: **ReconAgent/Project2**. Doctrine reference: **docs/updated_workbook.xlsx**. For local dev without scan binaries, set **TOOL MOCK MODE=true** in .env.

How you implement tool selection (ReAct, LLM tool-calling, rules) is your choice. The API must stay **prompt in → response out**.

7. Deliverables & Deadlines

#	Deliverable	Due
1	POST /agents/{id}/tasks working (prompt → response)	Week 1
2	Tool allowlist enforced	Week 1
3	GET /health endpoint	Week 1

#	Deliverable	Due
4	Team README (setup, env vars, curl example)	Week 2
5	3-minute demo: POST one prompt, show response	Week 2
6	Optional: pipeline demo with context chaining	Week 3

8. Grading (100 points)

Criteria	Points
API: POST prompt, structured response	30
Correct tool allowlist	20
Response quality on sample prompts	25
README + error handling	15
Demo	10

Bonus (+10): Striker team implements mock HITL gate; any team adds async task polling (final response only, no steps).

9. Rules & Safety

- Scan only **instructor-approved lab targets** or use mock mode.
- **Never** scan production systems without written approval.
- Team Striker must implement mock HITL approval before exploit tools run.
- Both teammates share ownership of the agent — split work fairly.
- Submit one shared repo folder or branch per team as directed by your instructor.

Questions? Contact your ReconAgent instructor before the Week 1 deadline.
Entersoft ReconAgent Program — Intern Assignment